Kno2®

# Building the Foundation

## The Financial Case for Universal Connectivity

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# Connecting Providers Across the Ecosystem

**Universal connectivity among all healthcare providers delivers substantial financial benefits, generating billions in direct cost savings and unlocking additional value through improved provider satisfaction, enhanced patient outcomes, and operational efficiencies.** Beyond these tangible gains, it paves the way for transformative digital innovation—enabling the U.S. healthcare system to overcome foundational challenges and adopt advanced care delivery models. These models integrate artificial intelligence, virtual care, and digital health solutions, marking a pivotal shift toward a more connected, efficient, and patient-centered future.

**Most healthcare professionals outside of health systems and physicians are not connected today**. This is for many reasons, including lack of incentives and requirements to connect.

- Skilled Nursing
- Home Health & Home Care
- Palliative & Hospice
- Physical & Occupational Therapy
- Behavioral Health
- Dental
- Vision
- Emergency Medical Services

Although there are billions in savings across all, Kno2 has selected the use cases from providers that are the most often overlooked:

- Physical Therapy
- Vision
- Dental
- Behavioral

# Why Include Physical Therapy

Physical therapy is fundamental to overall healthcare and value-based care because it provides non-invasive, cost-effective interventions that prevent expensive surgeries, reduce opioid dependence, and restore functional independence. Early physical therapy intervention can reduce total healthcare costs by 20-50% for musculoskeletal conditions while significantly decreasing hospital readmissions and recovery times. For chronic diseases like diabetes, heart disease, and COPD, physical therapy improves cardiovascular fitness and prevents costly complications like falls and fractures, particularly in aging populations.

From a value-based perspective, physical therapy addresses both treatment and prevention by maintaining patient mobility, reducing disability, improving workplace productivity, and enabling longer independence, ultimately keeping patients out of expensive acute care settings. When properly integrated into comprehensive care delivery, physical therapy serves as preventive medicine that delivers exceptional ROI while achieving the improved outcomes and reduced costs essential to value-based healthcare success.

**Total Therapists:**

- **Physical Therapists:** 240,000-250,000
- **Occupational Therapists: 1**50,000-180,000
- **Therapy Assistants:** 145,000-150,000

# Therapy Information Exchange Example:

## Therapy Professional

**Access from Medical Provider:**

- Diagnostic information- imaging results, surgical reports, lab value, differential diagnosis
- Current medications
- Physiological parameters
- Comorbidities – (i.e. Diabetes, osteoporosis, cardiac conditions)
- Recent medical changes – new symptoms, procedures

## Medical Provider

**Access from Therapy Professional:**

- Functional assessments – range of motion, strength, balance, standard outcome scores
- Progress tracking
- Pain patterns and activity responses
- Safety observations and fall risk assessments
- Treatment tolerance and adverse reactions

# Physical Therapy – MSK Early Intervention (Low Back Pain) Use Case

MSK conditions now exceed $380 billion annually, comprising 5.2% of the U.S. gross domestic product ($796.3 billion), representing the single largest health category and fastest growing healthcare cost in the United States. MSK conditions include back and neck pain, repetitive strain injuries, muscle strains and ligament pains, tendonitis, osteoporosis, fibromyalgia and more. Less than 10% of all Americans use therapy due to lack of referral and early coordination by primary and specialty care providers.

## Current National Financial Impact:

- **Direct MSK costs**: $380-420 billion annually
- **Percentage of GDP**: 5.2% of annual gross domestic product ($796.3 billion total economic impact)
- **Healthcare spending rank**: Single largest health category in the United States and fastest growing

## Low Back Pain – Early PT Interventions

- **Current approach**: Surgery-first averages $16,195 per episode
- **PT-first approach**: Averages $3,992 per episode
- **Savings per episode**: $12,203 (75% reduction)
- **Annual potential**: $29-44 billion saved on low back pain

## Why Universal connectivity

- Integration of data and services between primary care providers and therapists have been shown to significantly impact positive outcome's for MSK conditions including :
- Direct referrals and Direct PT access options
- Shared treatment plans
- Outcome tracking

# Why Include Vision Providers

Eye care is fundamental to overall healthcare and value-based care because vision problems often serve as early indicators of systemic conditions like diabetes, hypertension, and cardiovascular disease, making regular eye exams critical for preventive medicine and early intervention. For diabetic patients specifically, timely eye care coordination can prevent blindness—a catastrophic outcome that not only devastates quality of life but also triggers cascading healthcare costs including vision rehabilitation, increased fall risks, depression treatment, and lost productivity.

Universally connecting optometrists and incorporating their participation in co-managing patients assists in addressing our overall national provider/physician shortage.  Optometrists often see patients more regularly than primary care physicians for routine eye care and can detect systemic diabetes complications through comprehensive eye exams and identify non-compliance through retinopathy progression patterns.

**Total Eye Care Professionals:**

- Optometrists: 41,000-47,000
- Ophthalmologists: 17,000-18,000

# Vision Information Exchange Example:

## Eye Care Provider

**Access from Medical Provider to correlate to retinopathy findings:**

- Labs with most recent A1C results
- Blood pressure readings
- Medication lists
- Recent hospitalizations or diabetic complications
- Patient-reported diabetes statuses

## Medical Provider

**Access from Eye Care Provider:**

- Detailed retinopathy staging and progress rates
- Vision function assessments affecting self care
- Patients compliance with eye care recommendations
- Urgent findings requiring diabetes change management

# Vision Use Case – Diabetic Retinopathy

**Diabetic retinopathy** is a serious eye condition that affects people with diabetes. It occurs when high blood sugar levels damage the tiny blood vessels in the retina—the light-sensitive tissue at the back of the eye responsible for vision.

## Current National Financial Impact:

- **37 million Americans** with diabetes
- **11.7 million** with diabetic retinopathy
- **Current 5-year cost projection**: $2.8 billion

## Potential Connectivity Impact:

- **50% reduction** in preventable diabetic blindness
- **5-year savings**: $1.4 billion
- **Medicare portion**: $850 million
- **Break-even on connectivity costs** by Year 2-3

# Why Include Dental Providers

Dental care is fundamental to overall healthcare and value-based care because oral health serves as both a gateway and a mirror to systemic health conditions. Poor dental health is strongly linked to serious chronic diseases including diabetes, cardiovascular disease, stroke, and respiratory infections, with periodontal disease acting as both a risk factor and complicator for these conditions. For patients with diabetes, untreated gum disease can make blood sugar control significantly more difficult, while cardiovascular patients face increased risk of heart complications from oral bacteria entering the bloodstream.

Dental professionals often serve as the first line of detection for serious conditions like diabetes, cardiovascular disease, osteoporosis, and autoimmune disorders through oral manifestations, but without interoperability, these critical findings may never reach the patient's primary care physician or specialist. For diabetic patients, coordinated care between dental and medical providers is essential since periodontal disease can significantly impact blood sugar control, while uncontrolled diabetes increases infection risk and delays healing after dental procedures.

**Total Dental Professionals:**

- Dentists: 202,000
- Orthodontists: 10,904
- Endodontists: 5,000
- Periodontists: 6,000
- Hygienists: 215,000
- Assistants: 364,000

# Dental Information Exchange Example:

## Dental Professional

**Access from Medical Provider:**

- Complete medication lists
- Current conditions that may impact treatment planning
- Recent hospitalizations or patient reported conditions

## Medical Provider

**Access from Dental Professional:**

- Medications prescribed for oral care
- Detected problem list including potential sources of sepsis
- Assess anticoagulant interactions
- Understand recent procedures that may complicate treatment
- Recognize oral manifestations of systemic disease and could diagnosis

# Two Dental Use Cases

The impact of dental care has widespread implication and costs to the US healthcare system. Below are two identified use cases – 1) Periodontal diseases and diabetes 2) Medication Safety

## Current National Financial Impact:

- **$45 billion annually** in dental care costs in the U.S.
- **$15 billion in emergency department visits** for preventable dental conditions
- **$3.4 billion in lost productivity** from dental-related absences
- **Medical complications from untreated dental disease**: $6-8 billion annually

## Periodontal disease and diabetes:

- **Diabetic patients with gum disease** have $2,840 higher annual medical costs
- **22% of diabetic patients** have severe periodontal disease
- **Medical cost reduction** of 10-40% when dental disease is treated in diabetics
- **Total potential savings**: $2.5 billion annually for diabetic population

## Medication Safety & Adherence:

- **Dental prescriptions** (antibiotics, pain medications) interact with medical treatments
- **Adverse drug events** from uncoordinated prescribing: $3.5 billion annually
- **50% reduction potential** with connected prescribing systems

# Why Include Behavioral Providers

Behavioral care is fundamental to overall healthcare and value-based care because mental health conditions affect over 50% of Americans and significantly worsen medical outcome, depression doubles heart disease risk, anxiety increases emergency visits by 300%, and untreated mental health drives billions in preventable hospitalizations.

From a value-based perspective, behavioral health integration delivers exceptional ROI: every dollar invested saves $4 in reduced medical costs, while integrated models can reduce total healthcare costs by 20-30% through improved medication adherence and better chronic disease management. Mental health is also the leading cause of disability, with depression alone costing over $200 billion annually in lost productivity. When properly integrated, behavioral care enables early intervention, reduces costly crisis situations, improves patient engagement across all conditions, and addresses social determinants of health that drive poor outcomes, making it essential for achieving the improved outcomes and reduced costs that define successful value-based healthcare delivery.

**Total Behavioral Professionals & Providers:**

- Counselors, Therapists – 416,000
- Social Workers – 708,000
- Psychologists – 182,000
- Psychiatrists – 25,500

# Behavioral Information Exchange Example:

## Behavioral Professional

**Access from Medical Provider:**

- Complete medication lists
- Current conditions that may impact treatment planning
- Recent hospitalizations or patient reported conditions
- Laboratory results

## Medical Provider

**Access from Behavioral Professional:**

- Medications prescribed for mental health conditions
- Diagnosis
- Treatment plans, responses and outcomes
- Risk factors and safety information
- Social determinants and functional status
- Outcome measures

# Two Behavioral Use Cases

Mental health conditions are inherently integrated with physical health, requiring seamless coordination across the entire healthcare ecosystem to prevent dangerous medication interactions, identify underlying medical causes of psychiatric symptoms, and manage the complex interplay between mental and physical health conditions.

## Current National Financial Impact:

- **$280 billion annually** in total behavioral health care costs in the U.S.
- **$193 billion in lost productivity** from untreated mental health conditions
- **$147 billion in emergency department visits** for behavioral health crises
- **Medical complications from untreated behavioral health**: $100+ billion annually

## Depression and Chronic Disease:

- **Patients with depression and diabetes** have 50-75% higher medical costs
- **Additional annual medical costs**: $3,000-$5,000 per patient
- **26 million adults** with depression and chronic medical conditions
- **Total excess medical costs**: $78-130 billion annually
- **Coordination savings potential**: $40-65 billion annually

## Medication Safety & Adherence:

- **Psychiatric medications** interact with 60% of medical treatments
- **Medication non-adherence** costs $100-300 billion annually in medical complications
- **Coordinated prescribing** could prevent 30-50% of adverse events
- **Potential savings**: $30-150 billion annually